Assignment 7.2: Cooling system fault diagnosis

Scenario

A customer brings in a 2016 Ford Focus 1.0 EcoBoost. She says the temperature gauge has been creeping up on the motorway and she has had to top up the coolant twice in the last month. There are no puddles under the car.

During your inspection you recorded:

- Cooling system pressure test at 1.2 bar: pressure dropped to 0.6 bar in 10 minutes, no visible external leaks
- Combustion leak (block) test: test fluid changed from blue to yellow
- Some white smoke with a sweet smell from the exhaust after start-up
- Engine oil level and colour normal
- Coolant concentration measured with a refractometer: 30 % antifreeze

Tasks

Task 1. Describe the function of each of these components: radiator, thermostat, water pump, pressure cap, expansion tank, cooling fan. A table is acceptable.

Task 2. Describe the diagnostic procedure you carried out, in a logical order, including the PPE and safety precautions for working on a cooling system.

Task 3. Using the readings above, give your diagnosis and justify it. What repair would you recommend?

Task 4. The cooling system holds 7.0 litres. The coolant is at 30 % antifreeze and the manufacturer specifies 50 %. Calculate how much coolant must be drained and replaced with pure antifreeze. Show your working.

Task 5. Write a short explanation for the customer (non-technical) of what is wrong and why the car should not be driven until it is repaired.

Word count: 600 to 900 words. Submit as a Word document. Include your name and student number on the first page. Harvard referencing if you use sources.